

Seismic moment tensor event screening

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SUMMARY

Event screening is an explosion monitoring practice that aims to identify an event as an explosion ('screened in') or not ('screened out'). Confidence in event screening can be increased if multiple independent approaches are used. We describe a new approach to event screening using the seismic moment tensor and its representation on the hypersphere, specifically the 5-sphere of 6-degree unit vectors representing the normalized symmetric moment tensor. The sample of moment tensors from an explosion data set is unimodal on the 5-sphere and can be parametrized by the Langevin distribution, which is sometimes referred to as the Normal distribution on the hypersphere. Screening is then accomplished by finding the angle from the explosion population mean to any newly measured moment tensor and testing if that angle is in the tail of the Langevin distribution (conservatively quantified as greater than 99.9 per cent of the cumulative density). We apply the screen to a sample of earthquakes from the Western USA and the September 2017 explosion and subsequent collapse at the Punggye-Ri Test Site in North Korea. All the earthquakes and the collapse screen out, but the explosion does not.

Key words: Statistical methods; Earthquake monitoring and test-ban treaty verification; Earthquake source observations.

1 INTRODUCTION

The use of seismic data to determine the type of source (e.g. explosion, earthquake, collapse) that generated the waves is an important part of nuclear explosion monitoring. The calculation of the full seismic moment tensor (Minson & Dreger 2008) allows for discrimination of source-types based on the relative contribution of isotropic and deviatoric energy (Vasco 1990; Dreger & Woods 2002). Seismic moment tensors derived from regional full waveform fitting at intermediate periods (~ 10 – 100 s) show clear visual separation between explosion, earthquake and collapse sources using Hudson *et al.* (1989) source-type plots (Ford *et al.* 2009) or plotted on the fundamental lune of the eigensphere (Alvizuri *et al.* 2018). In this paper we describe an explosion population as a distribution on the unit hypersphere and develop a method to screen out non-explosion seismic events using their moment tensors.

Riedesel & Jordan (1989) offered a way to view the different components of a source, but their approach is more appropriate for analyses of single sources (with error) in a geographically oriented frame of the focal sphere, and is not ideal for discrimination of source populations. Whereas, source discrimination is a more general approach that deals with classification of all source-types, the more narrow approach of event screening has historically been used by explosion monitoring agencies (e.g. the International Data Centre of the Comprehensive Test Ban Treaty Organization). An event is screened out (or determined not to be an explosion) if the null hypothesis that the event is an explosion can be rejected (at some specified confidence). Therefore, the event screening approach requires the use of a statistical test based on the null hypothesis that the event is an explosion.

We set the screening framework in unit moment tensor space, which is the space of normalized moment tensors. The normalization dispenses with the scalar seismic moment and only contains information on relative moments of the force-couples. This space is appropriate for the standard event screening approach, in which event size is often a secondary consideration. The space of unit moment tensors has been extensively explored by Tape & Tape (2012) and later work, and our starting point is their formalization preliminaries. They point out the isomorphic mapping to the 5-sphere, as was done by Silver & Jordan (1982) and Willemann (1993), which is the space where we propose to perform seismic moment tensor screening.

Spherical statistics are covered in detail by Mardia & Jupp (1999) and Fisher *et al.* (1987) [and briefly, but importantly, in appendix B of Silver & Jordan (1982)], and they guide us in describing distributions on the hypersphere, where we will show that the observed explosion unit moment tensor population is not uniform and can be described by the Langevin distribution [or, as Silver & Jordan (1982) describe it, the hyperspherical normal distribution]. We estimate the distribution parameters and define a screening criterion based on the cumulative density

of the angular distance from the explosion population mean vector. We simulate the probability density of the estimated explosion population on the lune to gain some intuition and to compare with the uniform distribution. We also examine a sample of earthquakes for testing, and we use the proposed screen to the September 2017 explosion at the Punggye-Ri Test Site (PRTS) in the Democratic People's Republic of Korea (DPRK) and subsequent collapse.

2 PRELIMINARIES

Tape & Tape (2015) Sections 2 and 9 give us the motivation and formalism for using distributions on the hypersphere $\mathbb{S}^{p-1}$, where for a symmetric 3×3 moment tensor there are six unique elements, so $p = 6$, and we consider $\mathbb{S}^5$, or the 5-sphere, to describe unit moment tensor populations. We borrow from those sections here for consistency and ease of reference.

2.1 The space $\hat{\mathbb{M}}$ and hypersphere $\mathbb{S}^5$ of unit moment tensors

The euclidean inner product of 3×3 matrices $M = (m_{ij})$ and $N = (n_{ij})$ is defined by

$$M \cdot N = \sum_{i,j=1}^3 m_{ij} n_{ij}. \quad (1)$$

The norm of a matrix M is then defined in terms of the matrix inner product,

$$\|M\| = \sqrt{M \cdot M}. \quad (2)$$

This is defined as the Frobenius norm of the matrix, which is the same as the standard vector norm of a flattened 3×3 matrix. The space of all 3×3 symmetric matrices (moment tensors) of unit norm is denoted by $\hat{\mathbb{M}}$

$$\hat{\mathbb{M}} = \{M : M^T = M \text{ and } \|M\| = 1\}. \quad (3)$$

The 5-sphere, $\mathbb{S}^5$, is the set of points $\mathbf{v}$ in $\mathbb{R}^6$ with $\|\mathbf{v}\| = 1$. The function f defined by

$$f(x_1, \dots, x_6) = \begin{pmatrix} x_1 & x_6/\sqrt{2} & x_5/\sqrt{2} \\ x_6/\sqrt{2} & x_2 & x_4/\sqrt{2} \\ x_5/\sqrt{2} & x_4/\sqrt{2} & x_3 \end{pmatrix} \quad (4)$$

is a norm-preserving isomorphism (Silver & Jordan 1982) between $\mathbb{R}^6$ and the space of 3×3 symmetric matrices. In particular, f preserves inner products, $f(\mathbf{x}) \cdot f(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$, ($\mathbf{x}, \mathbf{y} \in \mathbb{R}^6$). The function f maps $\mathbb{S}^5$ onto $\hat{\mathbb{M}}$ of unit moment tensors. Willemann (1993) took advantage of this isomorphism to cluster unit moment tensors based on distance on the hypersphere. Here we will be use it to study the explosion population on the hypersphere.

3 DATA

We describe the explosion population using a sample of explosions: 18 explosions in the Western USA described by Ford *et al.* (2009) and five explosions in the DPRK described by Chiang *et al.* (2018); for a total of 23 events. The DPRK events in 2017 are excluded for testing in the last section. Fig. 1 shows these moment tensor solutions plotted on the fundamental lune of the eigensphere. An example representation of each of the moment tensor spaces described in the previous section is provided below for the underground nuclear explosion, AMARILLO (U.S. Department of Energy 2015), as measured by Ford *et al.* (2009). The seismic moment tensor m_{ij} is the size of the force couple in the direction of i along the j -axis,

$$\begin{pmatrix} m_{xx} & m_{xy} & m_{xz} \\ m_{yx} & m_{yy} & m_{yz} \\ m_{zx} & m_{zy} & m_{zz} \end{pmatrix} = \begin{pmatrix} 77.85 & -21.42 & 29.57 \\ -21.42 & 156.76 & 28.23 \\ 29.57 & 28.23 & 191.64 \end{pmatrix}$$

its normalized unit moment tensor $M \in \hat{\mathbb{M}}$ is

$$M = \begin{pmatrix} 0.291 & -0.080 & 0.110 \\ -0.080 & 0.586 & 0.105 \\ 0.110 & 0.105 & 0.716 \end{pmatrix}$$

and its mapping to a unit vector $\mathbf{x} \in \mathbb{S}^5$.

$$\mathbf{x}^T = (0.291, 0.586, 0.716, 0.149, 0.156, -0.113)$$

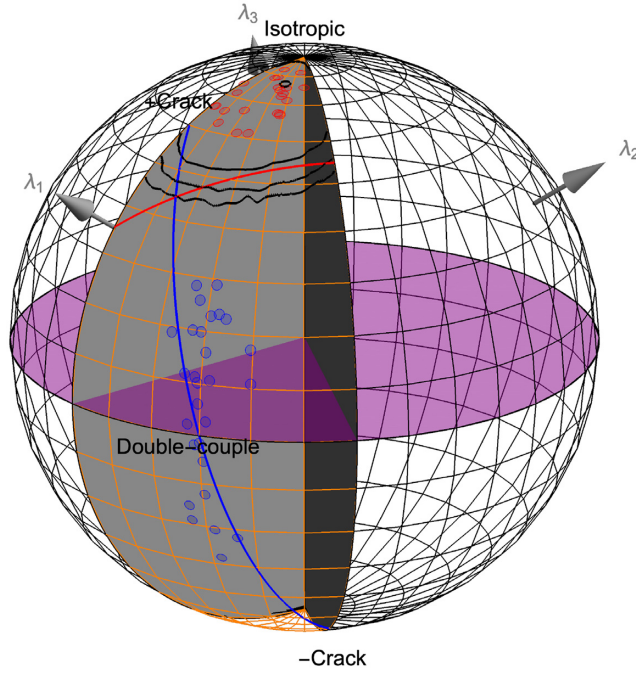


Figure 1. The fundamental lune of the eigensphere after Tape & Tape (2012), Fig. 1. The explosion sample are the red circles near the top of the lune and the sample mean vector (from $\mathbb{S}^5$) is the black circle. The earthquake sample are the blue circles near the midsection of lune. The black lines are the 0.01, 0.05 and 0.5 empirical density contours for a simulated sample from the Langevin distribution (in $\mathbb{S}^5$) that was estimated from the explosion sample.

4 SCREENING ON THE HYPERSPHERE

Distributions on the hypersphere are covered thoroughly in Mardia & Jupp (1999) and Watamori (1996), and we repeat many of their results here. We encourage the interested reader to reference those excellent texts for the thorough derivations. Similar derivations in a seismological context are in Silver & Jordan (1982, appendix B).

4.1 Descriptive measures

One can consider unit vectors in p -dimensional Euclidean space $\mathbb{R}^p$ as points on $\mathbb{S}^{p-1}$, the $(p-1)$ -dimensional sphere with unit radius and centre at the origin. Let $\mathbf{x}^\top = (x_1, \dots, x_p)$ and $\mathbf{x}_1, \dots, \mathbf{x}_n$ be points on $\mathbb{S}^{p-1}$. Then the location of these points can be summarized by their sample mean vector in $\mathbb{R}^p$, which is

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i. \quad (5)$$

For our explosion data set, $\bar{\mathbf{x}}^\top = (0.435, 0.506, 0.689, 0.0263, 0.0237, -0.109)$. It is useful to express it in polar form as $\bar{\mathbf{x}} = \bar{r} \bar{\mathbf{x}}_0$, where $\bar{\mathbf{x}}_0$ is a unit vector and $\bar{r} \geq 0$, so that $\bar{r} = \|\bar{\mathbf{x}}\|$ and $\bar{\mathbf{x}}_0 = \bar{\mathbf{x}} / \|\bar{\mathbf{x}}\|$. The unit vector $\bar{\mathbf{x}}_0$ is the sample mean direction and $\bar{r}$ is the mean resultant length. Thus $\bar{r}$ measures clustering around the mean direction. For our data set, $\bar{r} = 0.966$ and $\bar{\mathbf{x}}_0^\top = (0.450, 0.524, 0.713, 0.272, 0.0245, -0.112)$. In $\hat{\mathbb{M}}$, this is

$$\bar{\mathbf{M}} = \begin{pmatrix} 0.450 & -0.0794 & 0.0173 \\ -0.0794 & 0.524 & 0.0192 \\ 0.0173 & 0.0192 & 0.713 \end{pmatrix},$$

which is transformed to lune coordinates and plotted as a black circle in Fig. 1. If this were a focal mechanism plot (i.e. beachball diagram) it would be solid, since the first motions at all takeoff directions are positive, and the dominant axis would be almost vertical.

An important measure of dispersion is the scatter matrix $\bar{T}$ about the origin, defined by

$$\bar{T} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top. \quad (6)$$

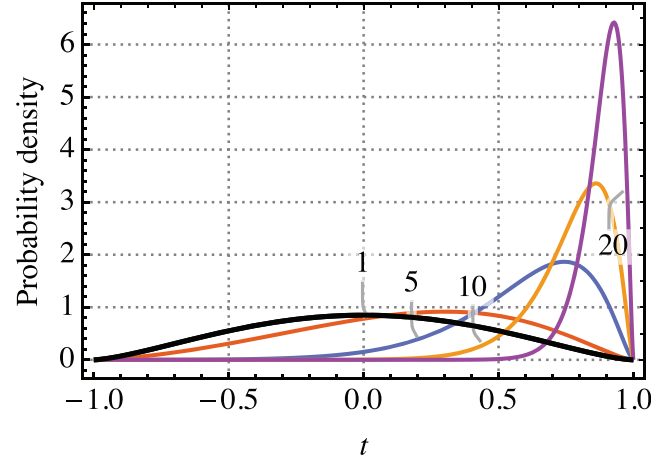


Figure 2. Marginal probability densities of t from the uniform distribution (thick black curve) as described by eq. (8) and the Langevin distribution $M_p(\mu, \kappa)$ as described by eq. (10) for a range of κ (callout value). As the concentration parameter κ increases the density at greater t increases.

For our data set,

$$\bar{T} = \begin{pmatrix} 0.200 & 0.227 & 0.287 & 0.00790 & 0.0109 & -0.0464 \\ 0.227 & 0.267 & 0.336 & 0.00826 & 0.0167 & -0.0551 \\ 0.287 & 0.336 & 0.495 & 0.0240 & 0.0135 & -0.0749 \\ 0.00790 & 0.00826 & 0.0240 & 0.0109 & 0.000165 & -0.00622 \\ 0.0109 & 0.0167 & 0.0135 & 0.000165 & 0.0107 & -0.00439 \\ -0.0464 & -0.0551 & -0.0749 & -0.00622 & -0.00439 & 0.0166 \end{pmatrix}$$

and the eigenvalues, $(\bar{t}_1, \dots, \bar{t}_p)$ of $\bar{T}$ are (0.935, 0.0397, 0.0118, 0.00891, 0.00305, 0.00198). The fact that the greatest eigenvalue $\bar{t}_1$ and $\bar{r}$ are close to unity suggest the explosion data come from a concentrated unimodal distribution in $\mathbb{S}^5$ (Fisher *et al.* 1987). This is most probably due to the property that observed explosions have a positive isotropic component with near vertical alignment due to gravity and surface-effects. Later we will show that this is not the case for an earthquake sample.

4.2 Uniform distribution

A key idea in distribution theory and data analysis on spheres is the ‘tangent-normal’ decomposition: any unit vector $\mathbf{x}$ can be decomposed as

$$\mathbf{x} = t\boldsymbol{\mu} + \sqrt{1-t^2}\boldsymbol{\xi} \quad (7)$$

with $\boldsymbol{\xi}$ a unit tangent to $\mathbb{S}^{p-1}$ at $\boldsymbol{\mu}$. We shall show some marginal distributions for t in the next sections on hypersphere distributions. The most basic probability distribution on the hypersphere is the uniform distribution, in which the probability of a set is proportional to its area. The uniform distribution is the unique distribution which is invariant under rotation and reflection. Therefore, the mean direction $\boldsymbol{\mu}$ is not defined. The marginal probability density of t is

$$(1-t^2)^{(p-3)/2} B\left(\frac{p-1}{2}, \frac{1}{2}\right)^{-1} \quad \text{for } -1 \leq t \leq 1, \quad (8)$$

where B is the Beta function. The density for $p = 6$ is shown in Fig. 2 as a thick black curve. The uniform distribution can be obtained by conditioning the multivariate normal distribution. Let $\mathbf{v} \sim N(\boldsymbol{\mu}, \mathbf{I}_p)$, where $\mathbf{I}_p$ is the identity matrix of order p , then the conditional distribution of $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ is uniform on $\mathbb{S}^{p-1}$. This technique was also mentioned by Tape & Tape (2015), Section 9 (i).

4.3 Langevin distribution

A unit random vector $\mathbf{x}$ has the $(p-1)$ -dimensional Langevin distribution $M_p(\boldsymbol{\mu}, \kappa)$ if its probability density function is

$$f(\mathbf{x}; \boldsymbol{\mu}, \kappa) = \frac{\kappa^{p/2-1}}{(2\pi)^{p/2} I_{p/2-1}(\kappa)} \exp(\kappa \boldsymbol{\mu}^\top \mathbf{x}), \quad (9)$$

where $\kappa > 0$, $\|\boldsymbol{\mu}\| = 1$ and I_n denotes the modified Bessel function of the first kind and order n . The parameters $\boldsymbol{\mu}$ and κ are called the mean direction and the concentration parameter, respectively. The Langevin distribution for $p > 2$ is also called the Von Mises-Fisher distribution.

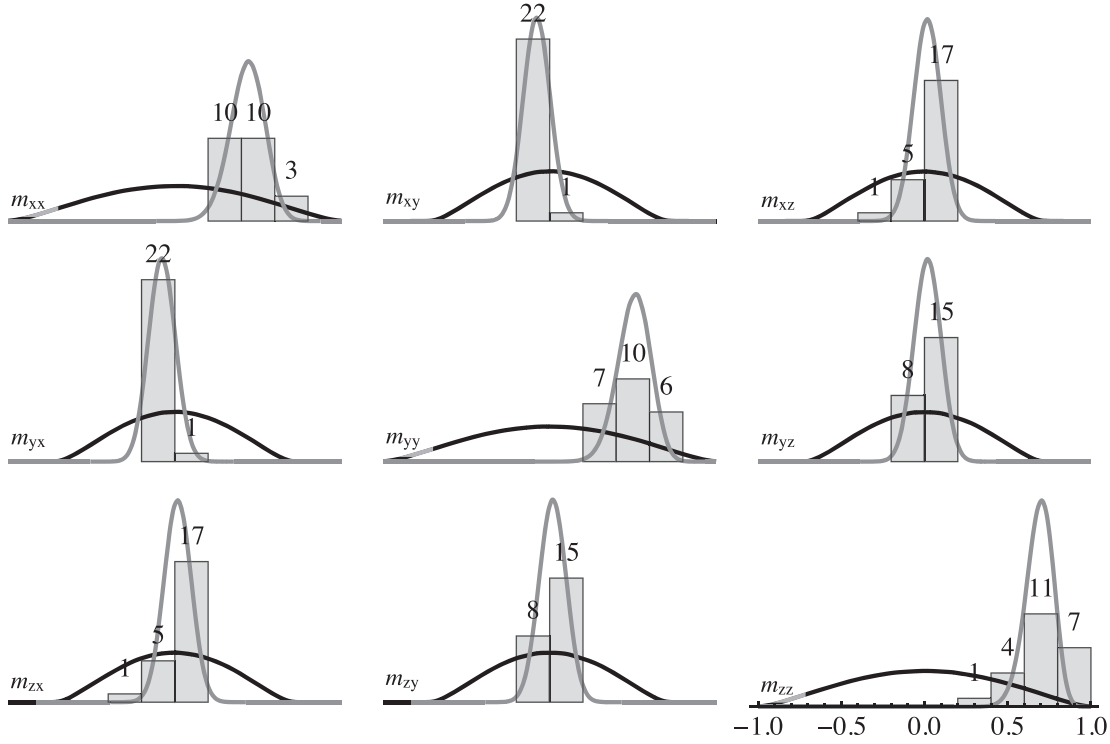


Figure 3. Comparison of moment tensor elements (labelled at left of each plot) from the explosion data set (histogram with counts per bin) and smoothed empirical density of elements simulated from a Langevin distribution (grey curve) parametrized by the explosion data set along with a uniform distribution (black curve) for comparison. The densities all integrate to one.

The marginal probability density of t in the tangent-normal decomposition is

$$\frac{\kappa^{p/2-1} e^{\kappa t} (1-t^2)^{(p-3)/2}}{\sqrt{\pi} 2^{p/2-1} \Gamma((p-1)/2) I_{p/2-1}(\kappa)} \text{ for } -1 \leq t \leq 1, \quad (10)$$

where Γ is the Gamma function. The density for $p = 6$ and a range of κ is shown in Fig. 2. Because $M_p(\mu, \kappa)$ is symmetric about μ , the mean direction of $\mathbf{x}$ is μ . Also, the larger the value of κ , the greater clustering around the mean direction. This behavior explains why κ is called the concentration parameter. Finally, Mardia & Jupp (1999) showed that the expectation of $\mathbf{x}$, $\mathbb{E}[\mathbf{x}]$, is a function of the mean direction and equal to $\rho\mu$ where

$$\rho = A_p(\kappa) = I_{p/2}(\kappa)/I_{p/2-1}(\kappa) \quad (11)$$

4.4 Parameter estimation

Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be a random sample from the Langevin distribution with probability density function as eq. (9). The vector mean is $\bar{\mathbf{x}}$. It follows from the general theory of exponential models that $\bar{\mathbf{x}}$ is sufficient to estimate κ and μ . The maximum likelihood estimate of the vector mean and concentration parameter is found by putting the population mean equal to the sample mean $\hat{\rho}\hat{\mu} = \bar{\mathbf{x}}$ so that $\hat{\rho} = \bar{r}$ and $\hat{\mu} = \bar{\mathbf{x}}_0$. Then $\bar{r} = A_p(\hat{\kappa})$ such that $\hat{\kappa}$ can be found from the inverse of the A_p function (eq. 11). Inference for the Langevin distribution is covered in Watamori (1996) and we refer the reader to that thorough manuscript for more details on parameter estimation.

For our data set then, we estimate $\hat{\mu}$ by $\bar{\mathbf{x}}_0$ from the explosion sample, and $\hat{\kappa}$ using the inverse formulation $A_6^{-1}(\bar{r})$ with $\bar{r}$ from the explosion sample, which gives

$$\hat{\mu}^\top = (0.450, 0.524, 0.713, 0.0272, 0.0245, -0.112) \text{ and } \hat{\kappa} = 73.7$$

4.5 Simulation

To gain some intuition for the estimate and comparison with the explosion data set, we simulate data from the estimated Langevin distribution, $M_6(\hat{\mu}, \hat{\kappa})$. If ξ and t [from eq. (7)] are generated independently from the uniform distribution on $\mathbb{S}^{p-1}$ and from the marginal distribution given in eq. (10), respectively, then $\mathbf{x} = t\mu + \sqrt{1-t^2}\xi$ is a pseudo-random unit vector with the Langevin distribution. We follow the algorithm described in Wood (1994), which is repeated in the Appendix. Fig. 3 shows a normalized histogram of the moment tensor elements from the explosion data set and the simulated elements from $M_6(\hat{\mu}, \hat{\kappa})$ and the uniform distribution (both in $\hat{\mathbb{M}}$).

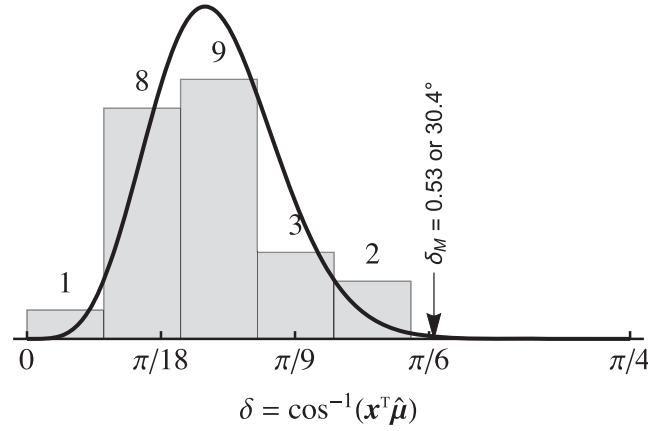


Figure 4. Angle between explosion population mean vector and explosion sample data set (histogram with counts per bin) and Langevin probability distribution (blue curve) described by eq. (13) along with angle at 99.9 per cent cumulative density, $\delta_M = 0.53$ or 30.4° , which is where we propose the screening criteria.

The explosion sample is better described by a Langevin distribution than a uniform distribution on the sphere. It also appears that the marginal distributions of each moment tensor element could be described by a normal distribution and therefore perhaps a multivariate normal distribution might be an appropriate description as was done by Riedesel & Jordan (1989). For large κ , as is the case here, the Langevin distribution does approach the Normal distribution. However, this choice of distribution could lead to unphysical simulations since the multivariate normal domain is unbounded and the normalized moment tensor elements are bounded to $-1 \leq m_{ij} \leq 1$ as in the Langevin distribution.

4.6 Screening criterion

The approach described here follows the general framework for magnitude-based screening as described in Fisk *et al.* (2002) and Selby *et al.* (2012). A screening statistic is usually defined as a distance from the explosion population mean to a newly estimated explosion parameter (e.g. m_b : M_s) weighted by the explosion population variance. The seismic moment tensor screening statistic is similar, but because we are on the hypersphere it is defined as the angle δ from the explosion population mean vector $\hat{\mu}$ to a newly estimated unit moment tensor $\mathbf{x}$ weighted by the explosion population concentration $\hat{\kappa}$. Fig. 4 shows δ , where

$$\delta = \cos^{-1}(\mathbf{x}^\top \hat{\mu}) \quad (12)$$

and $\mathbf{x}$ comes from the explosion sample and $\hat{\mu}$ was estimated from the explosion sample as described in the previous section. These observed angles are shown along with the probability distribution for δ ,

$$\frac{\kappa^2 e^{\kappa \cos(\delta)} \sin(\delta)^4}{3\pi I_2(\kappa)}, \quad (13)$$

which is transformed from the Langevin distribution for $p = 6$.

The modern approach uses the estimated screening statistic distribution to develop a screening criteria such that there is a low probability of screening an explosion out. Here, we propose the very conservative criteria that a newly observed moment tensor be within 99.9 per cent of the cumulative density to be screened in as a probable explosion (type-I error of 0.1 per cent). The angle at which this occurs δ_M is 0.531 radians or 30.4° and is shown in Fig. 4. We note that the greatest δ from the explosion sample belongs to the DPRK explosion on 9 September 2016, which is 0.447 radians or 25.6° .

4.7 Theoretical test

A nice feature of moment tensor screening is that one can test against canonical sources. Table 1 lists the angles for some of those sources. The angles to the canonical explosion and crack are less than δ_M , which means they would be screened in, and all others are greater and would be screened out. Another way to gain some intuition for the density of the explosion population distribution is to generate unit moment tensors from $M_6(\hat{\mu}, \hat{\kappa})$ and calculate the empirical density on the lune. Cumulative probability mass contours of 0.5, 0.95 and 0.99 are shown in Fig. 1.

5 APPLICATION TO WESTERN USA EARTHQUAKES

With an interest in developing an earthquake screening criteria, we analyse a set of 28 earthquake seismic moment tensors of events that occurred in the Western USA (Table 2). The data set of solutions was calculated using the approach described in Ichinose *et al.* (2003), which

Table 1. Angles δ for some canonical sources.

Name	M	$\mathbf{x}^\top$	$\cos^{-1}(\mathbf{x}^\top \hat{\boldsymbol{\mu}})$	$\delta [^\circ]$
Positive isotropic	$\frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\frac{1}{\sqrt{3}} (1, 1, 1, 0, 0, 0)$	0.227	13.03
Positive vertical crack	$\frac{1}{\sqrt{11}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$	$\frac{1}{\sqrt{11}} (1, 1, 3, 0, 0, 0)$	0.351	20.12
Positive vertical CLVD	$\frac{1}{\sqrt{6}} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$	$\frac{1}{\sqrt{6}} (-1, -1, 2, 0, 0, 0)$	1.385	79.35
45° dip-slip fault	$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\frac{1}{\sqrt{2}} (0, -1, 1, 0, 0, 0)$	1.436	82.29
Negative vertical crack	$\frac{1}{\sqrt{11}} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{pmatrix}$	$\frac{1}{\sqrt{11}} (-1, -1, -3, 0, 0, 0)$	2.790	159.88
Negative isotropic	$\frac{1}{\sqrt{3}} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\frac{1}{\sqrt{3}} (-1, -1, -1, 0, 0, 0)$	2.914	166.97

Table 2. Earthquake data set parameters.

Date and time	Latitude	Longitude	Depth	M_W	m_{xx}	m_{xy}	m_{xz}	m_{yy}	m_{yz}	m_{zz}
2006/02/16 17:47:59	37.981°	-118.775°	9 km	4.02	199	-45.5	434	513	-966	-185
2006/03/29 01:36:24	35.641°	-117.355°	18 km	3.69	11.1	-104	75.8	-184	-358	138
2006/05/05 06:36:19	38.228°	-118.757°	12 km	3.96	-39.3	280	220	-328	-105	1060
2006/05/29 10:38:43	39.366°	-120.465°	24 km	3.78	-136	194	-17.9	249	-194	425
2006/09/29 21:54:10	37.435°	-118.807°	9 km	3.51	64.8	28.6	-78.5	-20.4	122	133
2006/11/08 08:03:10	40.216°	-118.283°	9 km	3.55	-246	-18.1	3.44	-3.51	-98.5	84.4
2006/11/26 22:11:48	37.390°	-118.864°	9 km	3.84	85.3	15.9	-184	-260	436	479
2006/11/27 21:14:16	36.065°	-117.744°	12 km	3.90	143	247	-125	-166	-166	805
2006/12/15 19:21:58	36.246°	-120.001°	12 km	3.98	712	-339	-178	-1050	-175	-185
2006/12/16 06:14:05	36.153°	-120.226°	12 km	4.24	1800	-293	-215	-2530	-547	-798
2006/12/19 15:21:42	37.499°	-118.187°	3 km	3.60	-133	152	-45.2	-185	16.9	298
2007/01/17 14:01:08	37.400°	-118.542°	4 km	3.57	17.4	12.3	-12.3	-240	98.6	256
2007/01/24 11:30:16	37.413°	-117.099°	12 km	3.89	45.8	146	33.7	-2.04	-751	162
2007/02/23 00:25:16	40.151°	-116.233°	12 km	3.53	-119	32.6	64.3	214	-69.8	42.5
2007/02/28 11:47:41	41.154°	-114.870°	17 km	3.65	-80.9	-129	-36.2	-191	-238	196
2007/03/09 03:17:32	38.427°	-119.369°	6 km	4.75	-12200	3100	7910	-4480	-2820	5200
2007/03/30 09:09:32	35.880°	-117.839°	3 km	3.92	-335	212	214	181	-144	873
2007/04/27 15:42:50	37.120°	-115.292°	6 km	3.85	192	-103	-52.5	-141	-540	227
2007/04/27 19:55:11	37.131°	-115.294°	6 km	3.53	82.3	8.21	16.2	-50.9	-163	110
2007/06/11 01:03:46	37.466°	-113.997°	9 km	3.79	73.6	47.7	169	-196	-391	374
2007/06/12 07:22:36	37.534°	-118.863°	10 km	3.62	153	-34	-129	-170	-35.1	245
2007/06/12 07:23:43	37.533°	-118.874°	11 km	4.62	4080	-324	-3040	-4300	1490	9200
2007/07/29 04:57:46	40.550°	-115.931°	6 km	3.00	-36.9	-2.52	10.3	3.62	-12.5	-2.29
2007/08/06 05:59:44	37.766°	-114.334°	9 km	3.99	346	244	-381	685	-634	-226
2007/09/09 01:33:23	40.356°	-117.107°	4 km	3.09	-31.6	-8.14	-23.8	-1.58	-23.6	-4.73
2007/10/09 12:13:24	38.482°	-118.501°	6 km	3.86	57	288	2.59	-183	-143	698
2007/10/24 12:22:47	35.772°	-117.620°	6 km	4.03	-929	-510	-202	-267	-375	1310
2007/10/25 02:18:12	38.589°	-118.481°	3 km	3.32	-75.9	9.14	36.8	18.5	-28.8	102

Notes: All moment tensor elements (m_{ij}) are in μAk . The 2007 International Union of Geodesy and Geophysics/International Association of Seismology and Physics of the Earth's Interior (IUGG/IASPEI) General Assembly in Perugia, Italy, recommends 10^{18} N-m equal 1 Ak (Ak), so that 1 μAk is 1×10^{12} N-m.

uses regional intermediate period waveforms. The solutions were then set in the hyperspherical framework described above. Unfortunately, the eigenvalues of the scatter matrix $\tilde{T}$ are (0.367, 0.266, 0.189, 0.091, 0.063, 0.024), and the greatest eigenvalue from this sample is not close to one. This means that the sample and therefore the underlying population will not be well-described by an isotropic distribution on the hypersphere $\mathbb{S}^5$, such as the Langevin distribution. This is most probably due to the varying focal mechanisms in the data set and shows that the screening framework is sensitive to more than just source-type, but also the source frame or axes. Therefore, developing an earthquake screening criteria following the procedure used here for the explosion population is not feasible. However, we will use the earthquake data set as a further test of the explosion screen. Ichinose *et al.* (2003) estimate the error in their seismic moment tensor solutions to be about 10 per cent which would allow for confidence in any screening test.

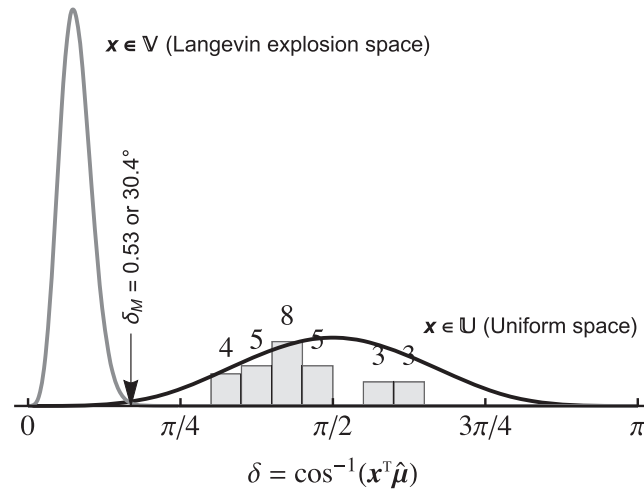


Figure 5. Density functions for the angles from $\hat{\mu}$ (estimated explosion population mean) for the set of moment tensors in the space defined by the estimated Langevin distribution (grey) and the Uniform distribution (black) and the earthquake sample (histogram with counts per bin).

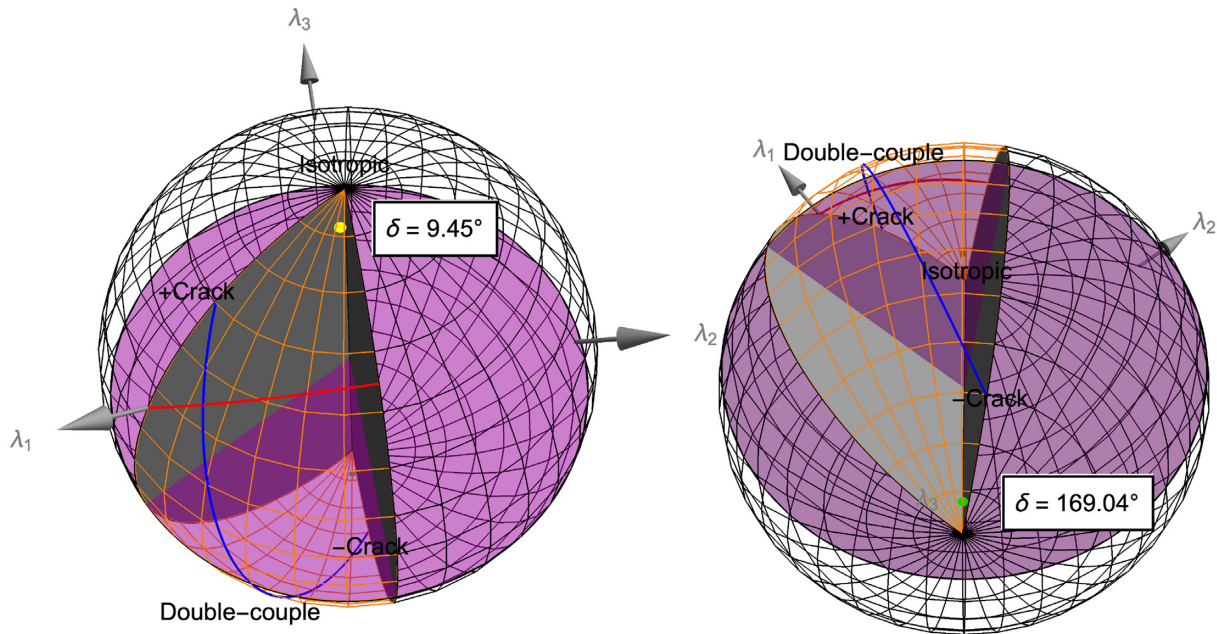


Figure 6. Test of screening criterion. The explosion (yellow) and collapse (green) near Mt Mantap in Sep 17 are plotted on the lune. The collapse is screened out since its screening angle is greater than $\delta_M = 30.4^\circ$.

Fig. 5 shows a histogram of the angles from the explosion population mean vector to the earthquake sample along with distributions of that angle for unit moment tensors from the Langevin explosion space $\mathbb{V}$ and the Uniform space $\mathbb{U}$. The distribution of angles from the Uniform space is $\frac{8}{3\pi} \sin(\delta)^4$, which has been explored by Tape & Tape (2016) and Tape & Tape (2017) in their description of a confidence parameter, and we are grateful to the authors for their illuminating discussions. Inspection of Fig. 5 reveals that none of the earthquakes have an angular distance less than 30.4° , so they are all screened out. In this case, there is no type-II error (screening in an earthquake), but a better test would include many more events; a task that is beyond the scope of this work and is available to future data wranglers.

6 TEST USING DPRK EVENTS

The event sequence in September 2017 at the PRTS near Mt Mantap in the DPRK provides a good test of source types for the seismic moment tensor screening approach. Regional moment tensors for an explosion and collapse were calculated by Chiang *et al.* (2018) and we screen them using the criterion proposed in the prior section. The solutions are plotted on the fundamental lune in Fig. 6 along with their δ . The collapse is screened out ($\delta > \delta_M$) and the explosion is not ($\delta \leq \delta_M$).

7 CONCLUSIONS

A screening criterion that uses the seismic moment tensor description of an event is developed using a sample of explosions and tested on a sample of earthquakes from the Western USA and an explosion and collapse in the DPRK. The set of normalized seismic moment tensors is analysed on the 5-degree hypersphere $\mathbb{S}^5$ and we find that it is well described by a unimodal distribution such as the Langevin distribution, or as Silver & Jordan (1982) describe it, the hyperspherical normal distribution. We infer the explosion population parameters, and the screening statistic is then the angle δ on $\mathbb{S}^5$ between the estimated explosion population mean $\hat{\mu}$ and a test event $\mathbf{x}$, or $\delta = \cos^{-1}(\mathbf{x}^\top \hat{\mu})$. The threshold angle δ_M of 30.4° is chosen such that 99.9 per cent of the cumulative density of the inferred Langevin distribution is less than δ_M (for a type-I error of 0.1 per cent).

In order to assess the goodness of fit for the inferred distribution, we use the simulation algorithm for a Langevin distribution from Wood (1994). The simulation algorithm performs well and shows that the Langevin distribution is an appropriate description for the population. Further, we note that the simulation algorithm could be used to empirically transform a random sampling from the Langevin distribution to other moment tensor parametrizations like the fundamental lune of Tape & Tape (2012).

We also analyse a sample of earthquakes from the Western USA and find it is not well suited for description by a unimodal distribution like the Langevin distribution, therefore we don't pursue a screening criteria as was done for the explosions. However, we do use the sample to test the explosion screening and find that all the 28 earthquakes screen out (type-II error for this small sample of 0 per cent). Finally, we test the explosion and collapse at the Punggye-Ri Test Site near Mt Mantap in the DPRK and find that the collapse ($\delta = 169^\circ$) is screened out (since $\delta_M < \delta$) and explosion ($\delta = 9.45^\circ$) is screened in (since $\delta_M > \delta$).

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SUPPORTING INFORMATION

Supplementary data are available at [GJI](https://doi.org/10.1093/gji/fgz007) online.

mtscreen.py.zip

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APPENDIX: SIMULATION ALGORITHM BY WOOD (1994)

Simulation of the Langevin distribution was described in an algorithm by Wood (1994) and is presented in Algorithm 1. For the simulations presented here, 1.5×10^6 generations resulted in 1060898 usable points (70.73 per cent). The simulated unit vectors are compared with the theoretical marginal distributions in Fig. A1.

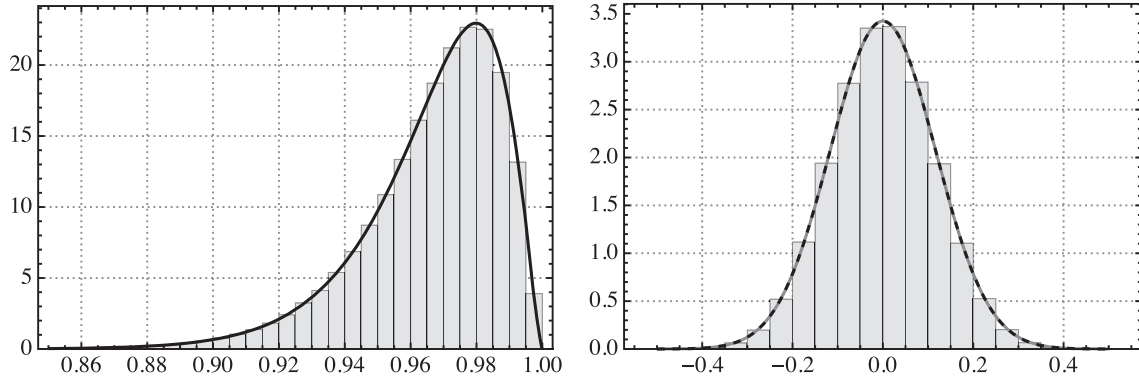


Figure A1. Histograms of w and $\sqrt{1-w^2}\mathbf{v}^\top$ from Algorithm 1, along with the probability densities (black curves) given by eq. (10) and $\sqrt{\kappa/2\pi}e^{-\kappa t^2/2}$, respectively. At high concentration, the conditioned distribution can be approximated by the limited Normal distribution with scale $1/\sqrt{\kappa}$ (dashed grey curve).

Algorithm 1 Algorithm VM* (Wood 1994)

```

1: procedure VM*( $\boldsymbol{\mu}, \kappa, p$ )
2:    $b \leftarrow -2\kappa/(p-1) + \sqrt{4\kappa^2 + (p-1)^2}/(p-1)$ 
3:    $x_0 \leftarrow (1-b)/(1+b)$ 
4:    $c \leftarrow \kappa x_0 + (p-1)\log(1-x_0^2)$ 
5:   Generate  $z \sim \text{Beta}[(p-1)/2, (p-1)/2]$  and  $u \sim \text{Uniform}[0, 1]$ 
6:    $w \leftarrow (1 - (1+b)z) / (1 - (1-b)z)$ 
7:   if  $\kappa w + (p-1)\log(1-x_0 w) - c \geq \log(u)$  then
8:     Create uniform  $(p-1)$ -dimensional unit vector,  $\mathbf{v}$ 
9:     Generate  $\hat{\mathbf{v}} \sim \text{MultivariateNormal}_{(p-1)}$ 
10:     $\mathbf{v} \leftarrow \hat{\mathbf{v}} / \|\hat{\mathbf{v}}\|$ 
11:     $\mathbf{x}^\top \leftarrow (\sqrt{1-w^2}\mathbf{v}^\top, w)$ 
12:  end if
13: end procedure

```

Then $\mathbf{x}$ has the Langevin distribution with modal direction $(0, \dots, 0, 1)^\top$ and concentration parameter $\kappa \geq 0$. In order to make $\mathbf{x} \sim M_p(\boldsymbol{\mu}, \kappa)$, find the rotation matrix that goes from $(0, \dots, 0, 1)^\top$ to $\boldsymbol{\mu}$ and apply it to $\mathbf{x}$.
